

@PersistenceContext VS @Autowired for EntityManager in Spring Boot



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Short Answer

`@PersistenceContext` is specifically designed for injecting `EntityManager` because `EntityManager` is **not a simple singleton bean** — it is a **stateful, non-thread-safe object** that needs a **separate instance per transaction or persistence context**.
`@Autowired` does not handle this lifecycle correctly.



What is a Persistence Context?

A **Persistence Context** is a set of entity instances that are being managed by an `EntityManager` at a given time. Think of it as a **first-level cache** and **unit of work** for your database operations.

Persistence Context

EntityManager manages:

- Which entities are "managed"
- Tracking changes (dirty checking)
- Flushing changes to DB
- First level cache

Lifecycle: tied to a TRANSACTION

◆ The Core Problem with @Autowired

```
// ❌ WRONG – Do NOT do this
@Repository
public class UserRepository {

    @Autowired
    private EntityManager entityManager; // DANGEROUS!
}
```

Why is this dangerous?

- EntityManager is **not thread-safe**
- If Spring injects a **single shared instance** via @Autowired , multiple threads could use the **same EntityManager simultaneously**
- This leads to:
 - **Data corruption**
 - **Unexpected behavior**
 - **Transaction mix-ups** between threads

◆ The Right Way — @PersistenceContext

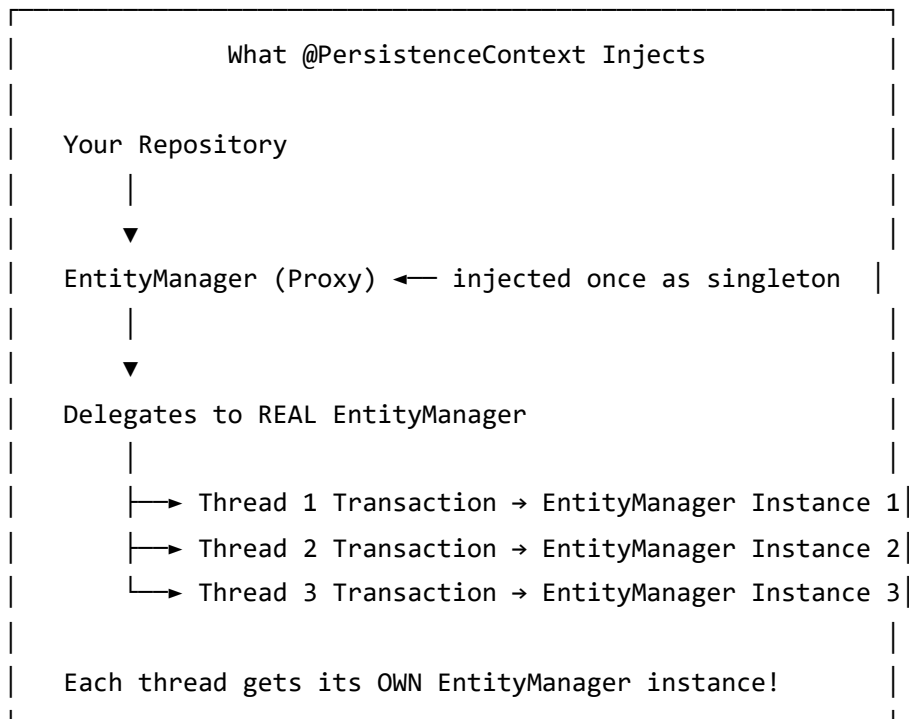
```
// ✅ CORRECT – Always use this
@Repository
public class UserRepository {

    @PersistenceContext
    private EntityManager entityManager; // SAFE!
}
```

Why is this safe?

Because `@PersistenceContext` does **NOT** inject a real `EntityManager` directly. Instead, it injects a **thread-safe proxy** of `EntityManager` .

◆ How the Proxy Works



So every time a **new transaction starts**, the proxy automatically routes to a **fresh EntityManager instance** bound to that transaction.

Code Comparison

Using @Autowired (Incorrect)

```
@Repository
public class UserRepository {

    @Autowired
    private EntityManager entityManager;
    // Single shared instance – NOT thread safe!
    // All transactions share same EntityManager
    // Can cause data leaks between transactions!

    public User findById(Long id) {
        return entityManager.find(User.class, id);
    }
}
```



Using @PersistenceContext (Correct)

@Repository

```
public class UserRepository {

    @PersistenceContext
    private EntityManager entityManager;
    // This is actually a PROXY
    // Each transaction gets its OWN EntityManager
    // Fully thread-safe!

    public User findById(Long id) {
        return entityManager.find(User.class, id);
    }

    public void save(User user) {
        entityManager.persist(user);
    }

    public User update(User user) {
        return entityManager.merge(user);
    }

    public void delete(Long id) {
        User user = entityManager.find(User.class, id);
        if (user != null) {
            entityManager.remove(user);
        }
    }
}
```



Types of Persistence Context

You can also define the **type** of persistence context with @PersistenceContext :

```
// 1. TRANSACTION scoped (DEFAULT)
// EntityManager lives as long as the transaction
@PersistenceContext(type = PersistenceContextType.TRANSACTION)
private EntityManager entityManager;

// 2. EXTENDED scoped
// EntityManager lives beyond a single transaction
// Used in Stateful EJBs or long-running conversations
@PersistenceContext(type = PersistenceContextType.EXTENDED)
private EntityManager entityManager;
```

Type	Lifecycle	Use Case
TRANSACTION <i>(default)</i>	Lives for one transaction only	Most common — typical CRUD operations
EXTENDED	Lives across multiple transactions	Long-running conversations, stateful components

◆ Key Differences Summary

Feature	@Autowired	@PersistenceContext
Designed for EntityManager	✗ No	✓ Yes
Thread Safety	✗ Not safe	✓ Safe via proxy
Injects Proxy	✗ No	✓ Yes
Transaction Awareness	✗ No	✓ Yes
New instance per transaction	✗ No	✓ Yes
JPA Standard	✗ Spring specific	✓ JPA standard annotation
Persistence Context Type support	✗ No	✓ Yes

◆ Does @Autowired Ever Work?

In **modern Spring Boot**, `@Autowired` on `EntityManager` may seem to work because Spring Boot also wraps it in a proxy automatically. But:

- It is **not guaranteed** behavior
- It is **not the standard** or recommended way
- It may **break** in non-Spring environments
- It gives you **no control** over persistence context type

// This might work in Spring Boot but is still BAD practice ❌

`@Autowired`

`private EntityManager entityManager;`

// Always prefer this ✅

`@PersistenceContext`

`private EntityManager entityManager;`

◆ Quick Summary

Why `@PersistenceContext`?

1. `EntityManager` is NOT thread-safe
2. `@PersistenceContext` injects a PROXY
3. Proxy gives each transaction its OWN EM
4. It is the JPA STANDARD annotation
5. Supports TRANSACTION & EXTENDED scope
6. `@Autowired` doesn't understand EM lifecycle

💡 Best Practice

Always use `@PersistenceContext` for `EntityManager` injection.
It is the **JPA standard**, **thread-safe**, and **transaction-aware** way

to work with EntityManager in any Spring or Java EE application! 🚀



References

- [Jakarta Persistence Specification](#)
- [Spring Framework Documentation — EntityManager](#)
- [Hibernate ORM Documentation](#)